$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \qquad K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$V = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \qquad d_k = 2 \qquad d_v = 2$$

$$QK^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$QK^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$\frac{QK^T}{\sqrt{d}} = \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 2/\sqrt{2} \end{bmatrix}$$

$$\frac{QK^T}{\sqrt{d}} = \begin{bmatrix} 0.707 & 0 & 0.707 \\ 0 & 0.707 & 0.707 \\ 0.707 & 0.707 & 1.414 \end{bmatrix}$$

$$\text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) = \begin{bmatrix} 0.401 & 0.197 & 0.401 \\ 0.197 & 0.401 & 0.401 \\ 0.248 & 0.248 & 0.503 \end{bmatrix}$$

$$\text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V = \begin{bmatrix} 0.401 & 0.197 & 0401 \\ 0.197 & 0.401 & 0.401 \\ 0.248 & 0.248 & 0.503 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.401+0+0.401 & 0+0.197+0.401 \\ 0.197+0+0.401 & 0+0.401+0.401 \\ 0.248+0+0.503 & 0+0.248+0.503 \end{bmatrix}$$

$$= \begin{bmatrix} 0.802 & 0.598 \\ 0.598 & 0.802 \\ 0.751 & 0.751 \end{bmatrix}$$

$$S_{masked} = S + M$$

$$= \begin{bmatrix} 0.707 & 0 & 0.707 \\ 0 & 0.707 & 0.707 \\ 0.707 & 0.707 & 1.414 \end{bmatrix} + \begin{bmatrix} 0 & -\infty & -\infty \\ 0 & 0 & -\infty \\ 0 & 0 & 0 \end{bmatrix}$$

$$S_{masked} = \begin{bmatrix} 0.707 & -\infty & -\infty \\ 0 & 0.707 & -\infty \\ 0.707 & 0.707 & 1.414 \end{bmatrix}$$

$$\text{softmax}(S_{masked}) = \begin{bmatrix} 1 & 0 & 0 \\ 0.330 & 0.670 & 0 \\ 0.248 & 0.248 & 0.504 \end{bmatrix}$$

$$\text{softmax}(S_{masked})V = \begin{bmatrix} 1 & 0 & 0 \\ 0.330 & 0.670 & 0 \\ 0.248 & 0.248 & 0.504 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0.330 & 0.670 \\ 0.752 & 0.752 \end{bmatrix}$$

A casual mask prevents each token from attending to future tokens. The scores corresponding to future positions are replaced with $-\infty$ before applying softmax. After softmax, these positions recieve zero attention weights. Now token 1 only attend to token 1. token 2 attend to token 1 & 2 token 3 attend to all three. This makes

sure that the model predicts the next token w/o attending future information.